

Space for questions/announcements.

- Quizzes, homework 1,2, post/pre examples/exercises in lecture notes.

- Textbook.

- maybe motivation on systems (more mathy)

$$x''' + x''x'x + x'x + xe^t = 0$$

$$y_1 = x$$

$$y_2 = x'$$

$$y_3 = x''$$

$$\rightarrow y_1' = x' = y_2$$

$$y_2' = x'' = y_3$$

$$y_3' = x''' = -x''x'x - x'x - xe^t \\ = -y_3y_2y_1 - y_2y_1 - y_1e^t$$

$$\begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ y_3' = -y_3y_2y_1 - y_2y_1 - y_1e^t \end{cases}$$

Lecture 05.

2.1, 2.2, 2.4

Systems: Modeling, Geometry and (easy) Analytical Methods.

Goal: Stay in the realm of what we know, i.e.,
1st order diff. equations.

Predator-prey

$x(t)$: prey population

$y(t)$: predator population

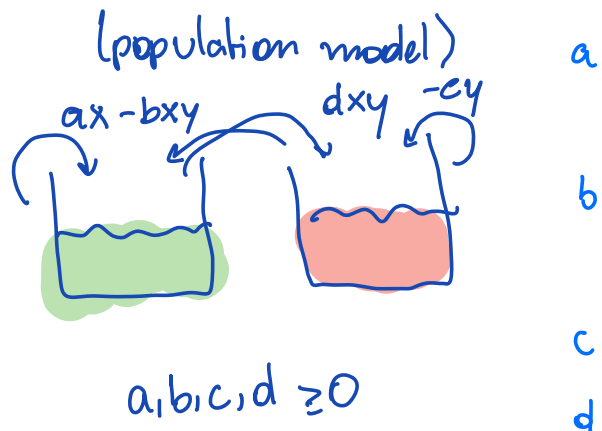
Factors involved:

- $x \rightarrow$ - ax

- $y \rightarrow x$ - $-bxy$

- $y \rightarrow$ - $-cy$

- $x \rightarrow y$ - dxy



Example: Suppose that a population of rabbit and foxes is modeled by

$$\frac{dR}{dt} = 0, \frac{dF}{dt} = 0$$

$$\begin{array}{l} a=3 \\ b=2 \\ c=2 \\ d=1 \end{array} \left\{ \begin{array}{l} \frac{dR}{dt} = 3R - 2RF = R(3-2F) = 0 \rightarrow R=0 \text{ or } F = \frac{3}{2} \\ \frac{dF}{dt} = -2F + RF = F(-2+R) = 0 \rightarrow F=0 \text{ or } R=2. \end{array} \right.$$

Natural questions:

a) $R(t)$ in the absence of $F(t)$?

$$R' = 3R \Rightarrow R(t) \rightarrow \infty \text{ as } t \rightarrow \infty$$

b) $F(t)$ in the absence of $R(t)$?

$$F' = -2F \Rightarrow F(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

c) Any constant solutions? $(0,0), (2, \frac{3}{2})$

d) Suppose $R(0) = 1$, $F(0) = 1$.

$$R' = 3R - 2RF$$

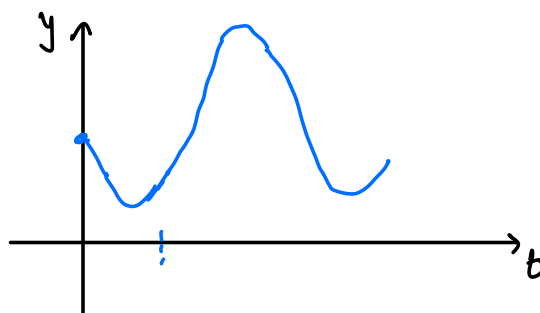
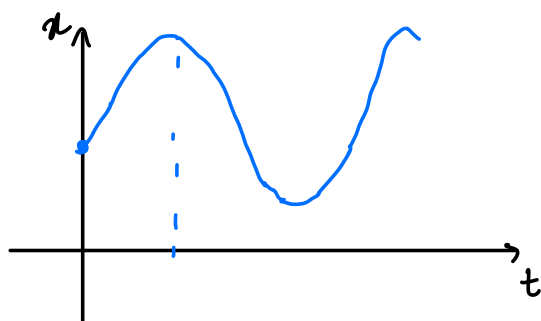
$$F' = -2F + RF$$

What happens in the long term?

$$R'(0) = 3 - 2 = 1$$

$$F'(0) = -2 + 1 = -1$$

⊕ plot $x(t), y(t)$



Motivation for next step:

$$\begin{cases} R' = 3R - 2RF \\ F' = -2F + RF \end{cases}$$

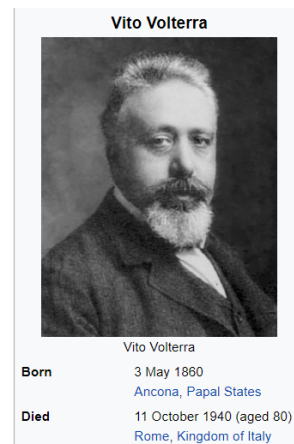
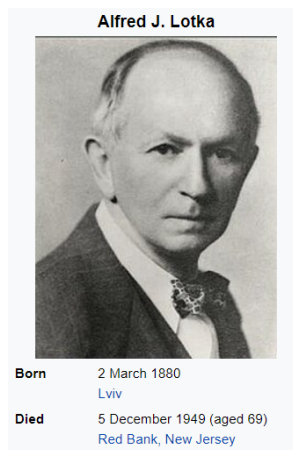
◦ What do we do when we don't know how to solve something?

A: We learn it.

Actually, prey-predator
don't really have nice closed
solutions. Any guess why?

↳ non-linear
terms.

(or Lotka-Volterra)



Qualitative Analysis:

Direction field

Ex: $x' = y$
 $y' = -x$

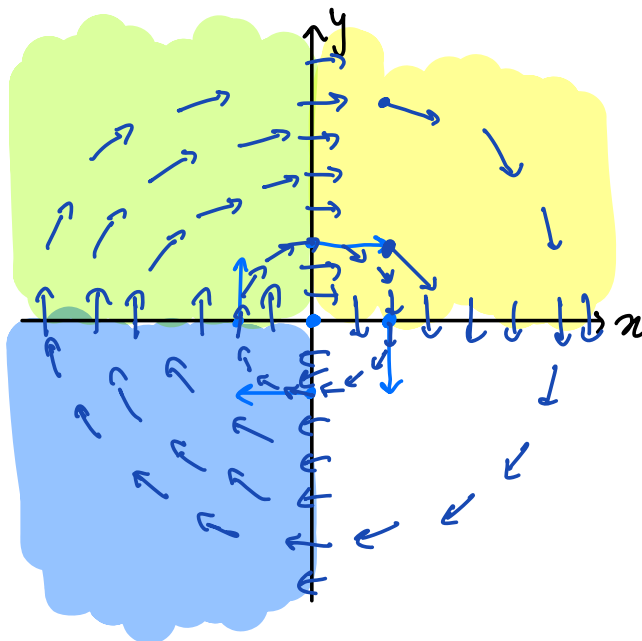
(x, y)	(x', y')
$(0, 0)$	$(0, 0)$
$(0, 1)$	$(1, 0)$
$(1, 0)$	$(0, -1)$
$(0, -1)$	$(-1, 0)$
$(-1, 0)$	$(0, 1)$

$x > 0, y > 0, x' > 0, y' < 0$

$x < 0, y > 0, x' < 0, y' < 0$

$x < 0, y < 0, x' < 0, y' > 0$

$x > 0, y < 0, x' > 0, y' > 0$



$$\begin{cases} x' = y \\ y' = -x \end{cases} \text{ is actually solvable!}$$

• What are the solutions?

$$x' = y$$

$$x'' = y' = -x$$

$$x'' = -x \leadsto \lambda^2 = -1 \Rightarrow \lambda = \pm i.$$

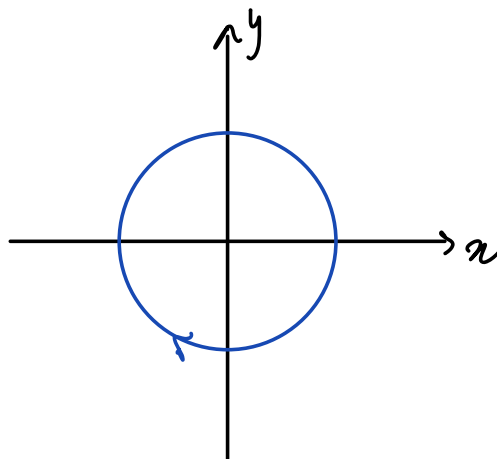
$$e^{it} = \cos(t) + i\sin(t)$$

$$\begin{aligned} x(t) &= c_1 \operatorname{Re}(e^{it}) + c_2 \operatorname{Im}(e^{it}) \\ &= c_1 \cos(t) + c_2 \sin(t). \end{aligned}$$

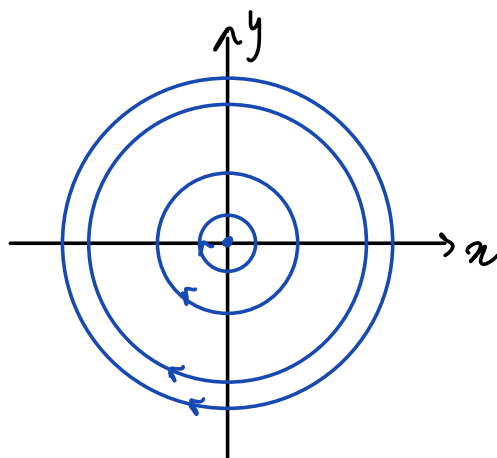
$$x(t) = \sin(t)$$

$$y(t) = \cos(t)$$

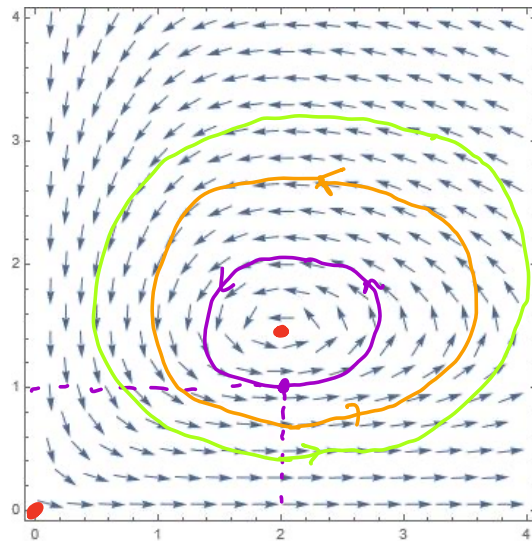
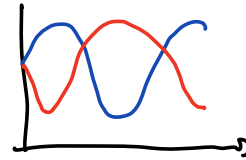
Phase - ^{line.} plane
(autonomous)



Phase - portrait



What about for the predator-prey?



a) Equilibria .

$(0,0), (2, \frac{3}{2})$

b) $x(0)=2, y(0)=1$.

logistic predator-prey model

$$\begin{cases} x' = 3x(1 - \frac{x}{5}) - 2xy \\ y' = -2y + xy \end{cases}$$

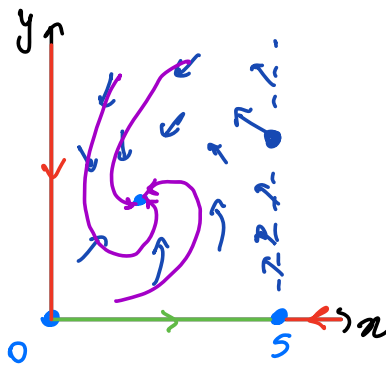
$$3 \cdot 5 \left(1 - \frac{5}{5}\right) - 2 \cdot 5 \cdot 0$$

$$0(-2 + 5)$$

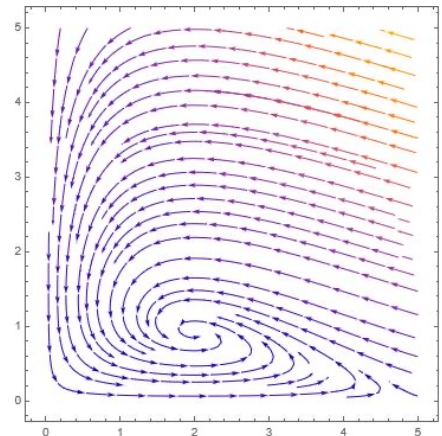
• Equilibrium solutions: $(0,0)$, $(5,0)$, $(2, \frac{9}{10})$

• Phase portrait:

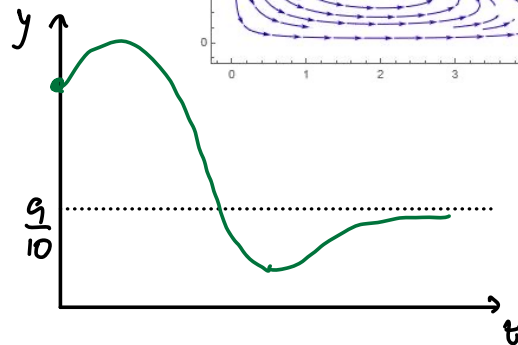
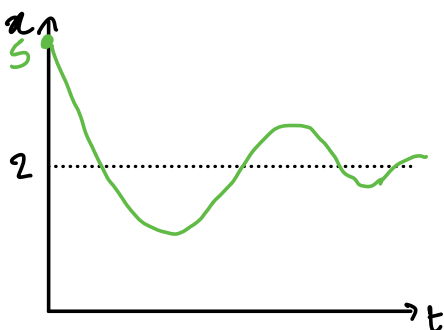
$y=0?$
 $x' = 3x(1 - \frac{x}{5})$
 $y'=0$
 $x=0?$
 $x'=0$
 $y' = -2y$



$x=5, y=1$
 $x' = -2 \cdot 5 \cdot 1 = -10$
 $y' = -2 + 5 = 3$
 $x=5, x' = -2y < 0$



solutions:

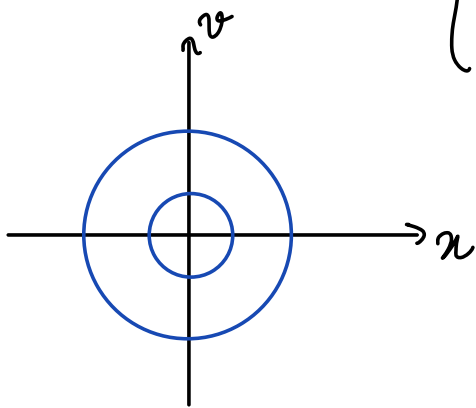


Mass-spring as a system.

$$m x'' + b x' + k x = 0$$

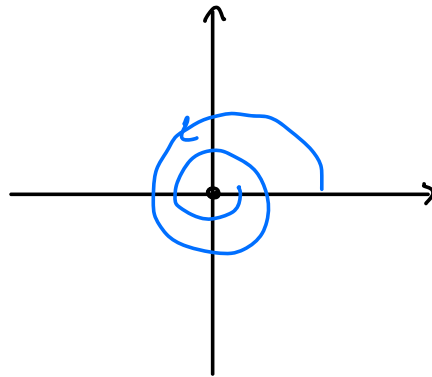
$$\begin{cases} x' = v \\ v' = -\frac{b}{m} v - \frac{k}{m} x \end{cases}$$

undamped: $b=0$



$$\begin{cases} x' = v \\ v' = -\frac{k}{m} x \end{cases}$$

underdamped: $b^2 - 4km < 0$



overdamp. / crit. damped

(matlab 3)

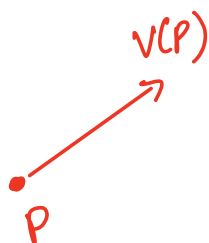
Geometry of systems. (vector notation)

$$\underline{\text{Ex.}} \quad \begin{cases} x' = x^2 + y^2 \\ y' = -2xy \end{cases}$$

def. vector field. $V(x(t), y(t)) = (x^2 + y^2, -2xy)$

$$\frac{d\vec{X}(t)}{dt} = V(\vec{X}(t))$$

• Direction field. $\leadsto$ Arrows = $(P, V(P))$, $P \in \mathbb{R}^2$.



Ex: Sketch solution curves and graph

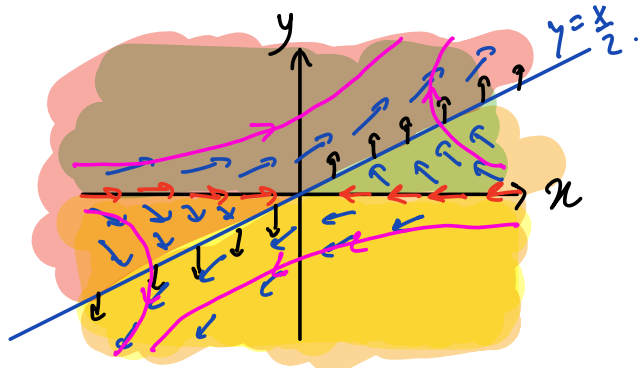
$$\begin{cases} x' = 2y - x \\ y' = y \end{cases}$$

$$\vec{V} = (2y - x, y)$$

$$\begin{array}{ll} 2y > x & y > \frac{x}{2} \Rightarrow x' > 0 \\ < x & y < \frac{x}{2} \Rightarrow x' < 0 \end{array}$$

Analysis of zeros and sign: $(0,0)$ is equilibrium.

$$\bullet V(x,y) = 0 \Rightarrow \begin{cases} x = 2y \\ y = 0 \end{cases} \Rightarrow x = 0$$

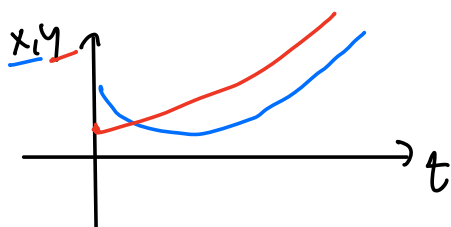


$$\begin{array}{ll} y' < 0 & y' > 0 \\ y = \frac{x}{2} \rightarrow y < \frac{x}{2} \Rightarrow x' < 0 & y > \frac{x}{2} \Rightarrow x' > 0 \end{array}$$

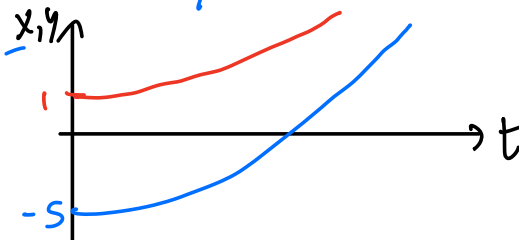
$$\begin{array}{ll} \text{at } y = \frac{x}{2}, x' = 0 & \uparrow \downarrow \\ \text{at } y = 0, y' = 0 & \rightarrow \leftarrow \end{array}$$

yellow + orange $\swarrow$
yellow + red $\searrow$
green + orange $\nwarrow$
green + red $\nearrow$

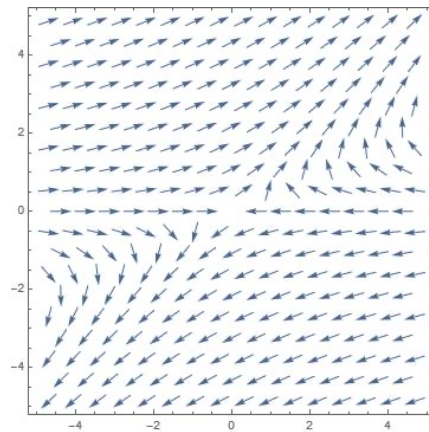
Sketch solution $x(0) = 5, y(0) = 1$



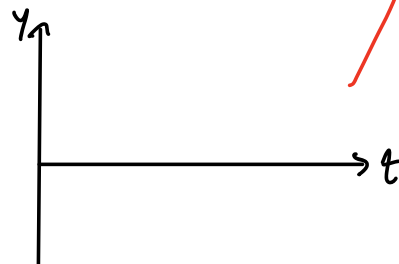
$x(0) = -5, y(0) = 1$



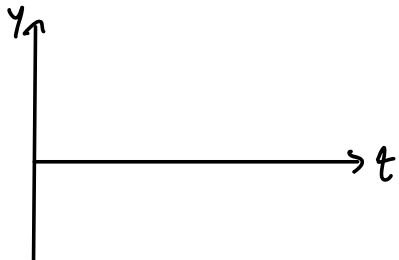
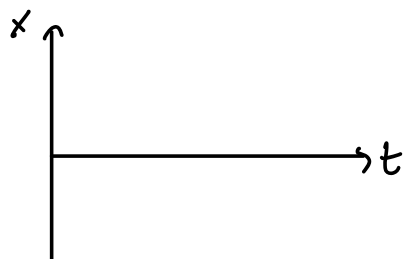
A proper direction field.



Task: sketch solutions. $(x_0, y_0) = (5, 1)$



Task: sketch solutions. $(x_0, y_0) = (-5, 1)$



here

Break if none yet.

General questions.

Analytical methods. via Examples :

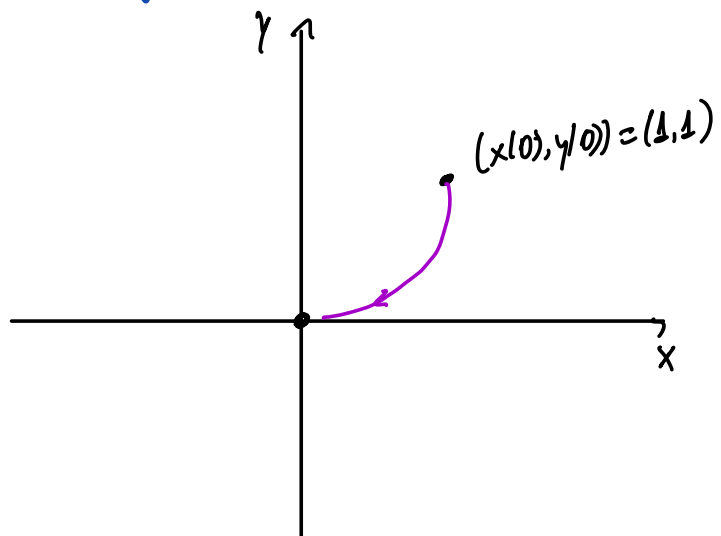
① Decoupled Systems.

$$\begin{cases} x' = -x \\ y' = -2y \\ x(0) = 1 \\ y(0) = 1 \end{cases}$$

$$\Rightarrow x(t) = D \cdot e^{-t}$$
$$\Rightarrow y(t) = C \cdot e^{-2t}$$

$$D = 1$$

$$C = 1.$$



② Partially decoupled.

$$\begin{cases} x' = 2y - x \\ y' = y \\ x(0) = 5 \\ y(0) = 1. \end{cases} \rightarrow y(t) = e^t$$

$$x' = -x + 2e^t$$

integrating factor
undetermined coeffs.

hw: find $x(t)$ with your favorite method.

③ Part. dec. + non-autonomous.

$$\begin{cases} x' = t x^2 \\ y' = \frac{1}{x^3} \\ x(1) = 1/2 \\ y(1) = 5/2 \end{cases}$$

$$x' = t x^2 \text{ (separable)}$$

$$\int \frac{dx}{x^2} = \int t dt \Rightarrow -\frac{1}{x} = \frac{t^2}{2} + C$$

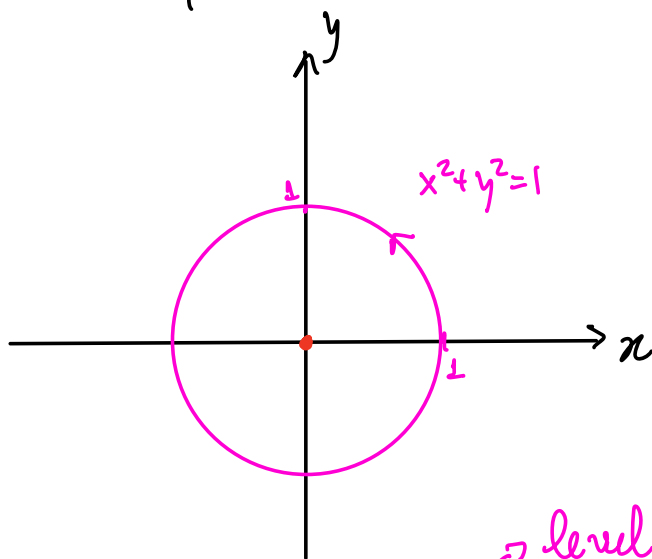
$$x(t) = \frac{-1}{\frac{t^2}{2} + C}$$

$$y' = \frac{-1}{(t^2 + C)^3}$$

Level sets of solutions

Ex $\begin{cases} x' = -y \\ y' = x \end{cases}$

$$V(x, y) = (-y, x)$$



$$H(x, y) = x^2 + y^2.$$

Question: What is the value of H along solutions of the SDE.

if $x(t), y(t)$ are solutions, what is

$$H(x(t), y(t)) \stackrel{?}{=} \begin{matrix} 0 \\ 1 \\ e^\pi + 1 \\ 42 \end{matrix}$$

→ level curves

Claim: $H(x(t), y(t)) = \text{constant}$. $\begin{matrix} x' = -y \\ y' = x \end{matrix}$

$H' = 0$, since $H(x, y) = x^2 + y^2$.

$$\frac{dH}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2x(-y) + 2y(x) = -2xy + 2xy = 0.$$

$\therefore H(x(t), y(t)) = \text{constant along solutions.}$

$$H(x, y) = 1.$$

if you can find such a $H(x, y)$, then you know where the solutions should be.

Predator - prey:

$$x' = 3x - 2xy$$

$$y' = -2y + xy$$

$$\frac{dy}{dx} = \frac{-2y + xy}{3x - 2xy} = \frac{y(-2+x)}{x(3-2y)}$$

$$\int \underbrace{\frac{(3-2y)}{y}}_{\frac{3}{y} - 2} dy = \int \underbrace{\frac{(-2+x)}{x}}_{-\frac{2}{x} + 1} dx$$

$$3 \ln|y| - 2y = -2 \ln|x| + x + C$$

$$3 \ln|y| + 2 \ln|x| - 2y + x = C.$$

$$H(x, y).$$

- Lecture: HW #2 (almost everything)

except: exist. + uniq
euler method